Title to be chosen

2026-11-13

Introduction

Plan

- **The gap.**
  - Real-time case counts are incomplete because of reporting delays.
  - Generic forecasting models do not use the reporting structure.
- **Context.**
  - Nowcasting methods model reporting delays directly.
  - Time series foundation models forecast well with no training on the target data.
  - Training on domain-specific synthetic data has precedent: KernelSynth for Chronos (Ansari et al. 2024), ForecastPFN (Dooley et al. 2023), prior-fitted networks (Müller et al. 2022; Hollmann et al. 2025), and Mantis for outbreaks (Dudley et al. 2026).
- **What we did.**
  - We fine-tuned a foundation model on synthetic outbreak data, following that precedent but fine-tuning a general model rather than training from scratch.
  - We evaluated it retrospectively and prospectively.

Methods

Data

> **Plan**
>
> - The German COVID-19 hospitalisation nowcast hub.
> - The Ebola forecasting hub target and its releases.

Baselines

> **Plan**
>
> - `randomwalk`, a random walk with no delay adjustment, and `randomwalknowcast`, a random walk nowcast.
> - Both in full in the SI.

Foundation model setup and fine-tuning

> **Plan**
>
> - Chronos-2, given the reporting triangle as series.
> - One fine-tune on synthetic data, run on a laptop.
> - Compared with Chronos-2 given only complete data points, to show what the ragged structure and fine-tuning add.

Synthetic data generation

> **Plan**
>
> - Simulated with epinowcast over a grid of outbreak, delay and noise settings.
> - A table in the main text gives the grid of synthetic scenarios.
> - The full process and an overview of the epinowcast model used are in the SI.

Table 1: Placeholder: grid of synthetic scenarios.

Setting	Values
Outbreak	

Setting	Values
Delay	
Noise	

Evaluation on held-out synthetic data

> **Plan**
>
> - A held-out set of simulations was not used during development.
> - Sample forecasts scored with CRPS and log-scale CRPS, bias, and coverage, relative to `randomwalknowcast`.
> - Scored against a rolling evaluation set of the data as reported D days later, with D each outbreak's 99% delay.

Case study 1: German nowcast hub

> **Plan**
>
> - Compared with the hub's models, relative to the hub's own baseline, and evaluated as the hub did.
> - Quantile forecasts scored with WIS and log-scale WIS, bias, and coverage, against a rolling evaluation set of the data as reported D days later (Wolffram et al. 2023).
> - Daily and weekly versions may not both make sense here.
> - The nowcast dates are split in half. We developed our setup on the first half only, and the second half is held out for the results.

Case study 2: prospective submission to the Ebola forecasting hub

> **Plan**
>
> - Compared with the other submissions, relative to the hub's own baseline.
> - Scored as for the held-out synthetic data, with D the empirical 99% point of the hub data's reporting delay.
> - Our models' forecasts are saved prospectively, alongside the main submissions as they were made.

Implementation

Plan

- The tools we used, each cited: Python, PyTorch on the laptop GPU, chronos-forecasting, PEFT for LoRA, R and epinowcast for simulation, and scoringrules for scoring, checked against scoringutils.
- The evaluation runs as a Snakemake workflow, so every result can be reproduced from the raw data.

Results

Plan

Each section has a figure panel. Each panel shows:

- nowcasts over time (ribbons);
- scores relative to the baseline (`randomwalknowcast` for synthetic data, the hub's baseline for each case study), with natural- and log-scale WIS (or CRPS, MAE where they make sense);
- bias;
- 50% and 90% coverage;
- score variation by location and horizon, supported by SI plots.
- for Germany, a dotted line marking the split between the development and held-out halves, with scores split by half.

The plots in primarycensored-paper are a model for these. Each section has two paragraphs, one on visual performance and one summarising scores.

Held-out synthetic data

Plan

- Visual performance of the candidate models against `randomwalk` and `randomwalknowcast`.
- A summary of score performance.

Figure to come.

Figure 1: Placeholder: held-out synthetic data results panel.

German nowcast hub

> **Plan**
>
> - Visual performance against the other models.
> - A summary of score performance.

Figure to come.

Figure 2: Placeholder: German nowcast hub results panel.

Ebola hub submissions

> **Plan**
>
> - Visual performance against the other submissions.
> - A summary of score performance.

Figure to come.

Figure 3: Placeholder: Ebola hub submissions results panel.

Discussion

> **Plan**
>
> - **Summary** of what we did.
> - **Strengths and weaknesses.**
>   - Weaknesses: only one foundation model, only one source of synthetic data, and limited case studies.
>   - Weakness: we did not compare against epinowcast, or, for the synthetic study, against the data-generating model itself.
>   - Strengths: both retrospective and real-time case studies, and synthetic data that are easy to tune to a setting.
>   - Strength: fine-tuning ran locally on consumer hardware (a laptop), so it is accessible.
>   - Weakness: fine-tuning used the laptop's GPU, which made it several times faster, and not everyone has one. On a laptop CPU alone it would still run in hours.
> - **Domain knowledge through synthetic data.**

  - Domain knowledge enters through the synthetic data source.
  - Fingers crossed it improves results; to fill in from the results.
- **Against the literature.**
  - Other uses of foundation models in infectious disease modelling.
  - Fine-tuning in other domains.
  - Other nowcasting approaches.
- **A possible emulator.**
  - Fine-tuning on epinowcast simulations may give a fast emulator of that package.
  - We did not compare against epinowcast, so we cannot say how close it is.
  - This could be useful for other slow, long-running packages.
- **Further work.**
  - Other foundation models.
  - More case studies.
  - A comparison with custom neural networks.
  - Testing the fine-tuned model as an emulator of epinowcast.
  - Fine-tuning with other epidemiological features as covariates, such as distances between locations or mobility.
- **Conclusions.**
  - Reuse state-of-the-art generic forecasting models with domain-specific insight.
  - Fine-tuning on a package's simulations could give a fast emulator of that package, which may help other long-running packages.
  - It shows promise, and more work is needed.

Funding

Acknowledgements

Data and software availability

AI disclosure

References

Ansari, Abdul Fatir, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, et al. 2024. "Chronos: Learning the Language of Time Series." arXiv. https://doi.org/10.48550/arXiv.2403.07815.

Dooley, Samuel, Gurnoor Singh Khurana, Chirag Mohapatra, Siddartha Naidu, and Colin White. 2023. "ForecastPFN: Synthetically-Trained Zero-Shot Forecasting." arXiv. https://doi.org/10.48550/arXiv.2311.01933.

Dudley, Carson, Reiden Magdaleno, Christopher Harding, Ananya Sharma, Emily Martin, and Marisa Eisenberg. 2026. "Mantis: A Foundation Model for Mechanistic Disease Forecasting." *Proceedings of the National Academy of Sciences* 123 (40). https://doi.org/10.1073/pnas.2602542123.

Hollmann, Noah, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. "Accurate Predictions on Small Data with a Tabular Foundation Model." *Nature* 637 (8045): 319–26. https://doi.org/10.1038/s41586-024-08328-6.

Müller, Samuel, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. 2022. "Transformers Can Do Bayesian Inference." In *International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2112.10510.

Wolffram, Daniel, Sam Abbott, Matthias an der Heiden, Sebastian Funk, Felix Günther, Davide Hailer, Stefan Heyder, et al. 2023. "Collaborative Nowcasting of COVID-19 Hospitalization Incidences in Germany." *PLOS Computational Biology* 19 (8): 1–25. https://doi.org/10.1371/journal.pcbi.1011394.